

Consider the following statements

1. If the relation is in 3NF, then it is also in 2NF
2. If the relation is in 3NF, then it is also in BCNF
3. If the relation is in BCNF, then it is also in 3NF

- ☐ Statements 1 & 2 are correct
- ☐ Statements 1 & 3 are correct
- ☐ All the statements are correct
- ☐ Only statement 1 is correct

Which among the following statements is correct?

2 points

- ☐ It is always possible to get a BCNF decomposition that is dependency preserving.
- ☐ A dependency $X \rightarrow Y$ is allowed in 3NF if Y is a super key or Y is a part of some key.
- ☐ A relation that is in BCNF, is also in 1NF, 2NF, 3NF and 4NF.
- ☐ None of the above

Consider relation $\mathbf{R} = (A, B, C, D)$ having functional dependencies $\{C \rightarrow D, C \rightarrow A, B \rightarrow C\}$.

2 points

Let $\mathbf{R}$ be decomposed into

$\mathbf{R1} = (A, B, C)$, $\mathbf{R2} = (C, D)$, $\mathbf{R3} = (C, A)$, and $\mathbf{R4} = (B, C)$. Then, which of following options is correct?

- ☐ $\mathbf{R2} = (C, D)$ and $\mathbf{R1} = (A, B, C)$ are in BCNF
- ☐ $\mathbf{R3} = (C, A)$, $\mathbf{R1} = (A, B, C)$ and $\mathbf{R4} = (B, C)$ are in BCNF
- ☐ $\mathbf{R2} = (C, D)$, $\mathbf{R3} = (C, A)$, and $\mathbf{R4} = (B, C)$ are in BCNF
- ☐ $\mathbf{R1} = (A, B, C)$ and $\mathbf{R4} = (B, C)$ are in BCNF

Consider the relation $\mathbf{R} = (A, B, C, D, E, H)$ having functional dependencies $F = \{A \rightarrow BC, E \rightarrow HA\}$. Let $\mathbf{R1} = (A, B, C)$ and $\mathbf{R2} = (A, D, E, H)$ be a decomposition of $\mathbf{R}$. Then, which

2 points

of the following is correct?

- ☐ Both $\mathbf{R1} = (A, B, C)$ and $\mathbf{R2} = (A, D, E, H)$ are in BCNF
- ☐ $\mathbf{R1} = (A, B, C)$ is in BCNF and $\mathbf{R2} = (A, D, E, H)$ is not in BCNF
- ☐ Both $\mathbf{R1} = (A, B, C)$ and $\mathbf{R2} = (A, D, E, H)$ are not in BCNF
- ☐ $\mathbf{R1} = (A, B, C)$ is not in BCNF and $\mathbf{R2} = (A, D, E, H)$ is in BCNF

Consider the relation $\mathbf{R} = (A, B, C)$ having functional dependencies $F = \{A \rightarrow B, B \rightarrow C\}$. **2 points**
If we decompose the relation $\mathbf{R}$ into $\mathbf{R1} = (B, C)$ and $\mathbf{R2} = (A, B)$ then, which of the following is/are correct?

- ☐ Relation $\mathbf{R}$ is in BCNF
- ☐ Relation $\mathbf{R1}$ and $\mathbf{R2}$ are in BCNF
- ☐ Relation $\mathbf{R1}$ and $\mathbf{R2}$ are not in 3NF
- ☐ None of the above

Which of the following statements is/are correct regarding the third normal form?

2 points

- ☐ It concentrates on Candidate Key
- ☐ Redundancy is high as compared to BCNF
- ☐ It concentrates on Primary key
- ☐ All the above

Which of the following statements is/are correct regarding Boyce–Codd Normal Form?

2 points

- ☐ It focuses on candidate keys
- ☐ Redundancy is high as compared to 3NF
- ☐ It preserves all the dependencies
- ☐ A dependency $X \rightarrow Y$ is allowed if X is a super key

If we decompose relation $\mathbf{R}(A, B, C, D, E, F, G, H)$ having functional dependencies $\{A \rightarrow B, ABCD \rightarrow E, EF \rightarrow GH, ACDF \rightarrow EG\}$ into

2 points

$\mathbf{R1}(A, B, C, D, F)$, $\mathbf{R2}(A, B)$, $\mathbf{R3}(A, B, E, F)$ and $\mathbf{R4}(E, F, G, H)$, then which of the following relations is/are in 3NF?

- ☐ $\mathbf{R1}(A, B, C, D, F)$
- ☐ $\mathbf{R2}(A, B)$
- ☐ $\mathbf{R3}(A, B, E, F)$
- ☐ $\mathbf{R4}(E, F, G, H)$

A relational schema $\mathbf{R}$ is said to be in 3NF if for every FD $X \rightarrow A$,

2 points

- ☐ X is a superkey of R
- ☐ A is a superkey of R
- ☐ A is part of some candidate key
- ☐ All of the above

If we decompose a relation into a set of relations which are in BCNF, then which of the following options is/are correct?

2 points

- ☐ The decomposition is lossless.
- ☐ It may not preserve the dependencies.
- ☐ Redundancy is high as compared to 3NF.
- ☐ The decomposition is lossy.